

# SQL DATA ANALYSIS REPORT

## Project 3 — SQL Data Analysis

DecodeLabs Internship | Batch 2026

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### 1. Introduction

This report presents the findings from an SQL Data Analysis conducted on an e-commerce dataset of 1,200 orders using Microsoft SQL Server Management Studio (SSMS). The goal was to extract meaningful business insights through structured SQL queries.

### 2. Queries Written & Results

```
5
6 SELECT OrderID, Product, Quantity, UnitPrice, TotalPrice
7 FROM EcommerceData
8
9 --- Query 2: Filter Deliverd orders
10 SELECT OrderID, Product, OrderStatus,
11 ROUND(TotalPrice, 2) AS TotalPrice
12 FROM EcommerceData
```

130 % No issues found Ln: 9, Ch: 1

|    | OrderID   | Product | Quantity | UnitPrice        | TotalPrice       |
|----|-----------|---------|----------|------------------|------------------|
| 1  | ORD200000 | Monitor | 5        | 570.619995117188 | 2853.10009765625 |
| 2  | ORD200001 | Phone   | 2        | 151.350006103516 | 302.700012207031 |
| 3  | ORD200002 | Tablet  | 5        | 550.679992675781 | 2753.39990234375 |
| 4  | ORD200003 | Chair   | 1        | 273.190002441406 | 273.190002441406 |
| 5  | ORD200004 | Printer | 4        | 626.010009765625 | 2504.0400390625  |
| 6  | ORD200005 | Phone   | 2        | 245.860000610352 | 491.720001220703 |
| 7  | ORD200006 | Laptop  | 1        | 664.419982910156 | 664.419982910156 |
| 8  | ORD200007 | Monitor | 5        | 149.550003051758 | 747.75           |
| 9  | ORD200008 | Phone   | 2        | 134.279998779297 | 268.559997558594 |
| 10 | ORD200009 | Desk    | 4        | 509.380004882813 | 2037.52001953125 |
| 11 | ORD200010 | Tablet  | 5        | 625.969970703125 | 3129.85009765625 |
| 12 | ORD200011 | Monitor | 2        | 40.1200000000000 | 80.2400000000000 |

Query 1 — Basic Data Retrieval

Retrieved key order details including OrderID, Product, Quantity, UnitPrice and TotalPrice from the dataset.

#### Query 2 — Filtered Delivered Orders

```
8
9      --- Query 2: Filter Delivered orders
10     SELECT OrderID, Product, OrderStatus,
11            ROUND(TotalPrice, 2) AS TotalPrice
12     FROM EcommerceData
13     WHERE OrderStatus = 'Delivered'
14
15     --- Query 3: How many orders does each pro
```

% No issues found

Results Messages

| OrderID   | Product | OrderStatus | TotalPrice |
|-----------|---------|-------------|------------|
| ORD200004 | Printer | Delivered   | 2504.04    |
| ORD200015 | Printer | Delivered   | 473.96     |
| ORD200021 | Monitor | Delivered   | 1371.8     |
| ORD200024 | Laptop  | Delivered   | 461.9      |
| ORD200025 | Monitor | Delivered   | 399.98     |
| ORD200026 | Laptop  | Delivered   | 297.75     |
| ORD200029 | Desk    | Delivered   | 1270.59    |
| ORD200032 | Tablet  | Delivered   | 2683.6     |
| ORD200038 | Laptop  | Delivered   | 245.89     |
| ORD200039 | Printer | Delivered   | 769.38     |
| ORD200040 | Desk    | Delivered   | 688.04     |

Filtered orders with status "Delivered" — revealing 231 successfully delivered orders out of 1,200 total.

#### Query 3 — Product Analysis

Printer was the most ordered product with 181 orders while Phone was least ordered with 156 orders.

#### Query 4 — Total Revenue

```
0
1  --- Query 4: Total Revenue from Orders
2  SELECT SUM(ROUND(TotalPrice, 2)) AS TotalRe
3  FROM EcommerceData
4
5  --- Query 5: What is the average amount spen
6  SELECT ROUND(AVG(TotalPrice), 2) AS Average
7  FROM EcommerceData
8
9  --- Query 6: Which product generates the mo
0  SELECT Product,
```

▼

✓ No issues found

Results Messages

| TotalRevenue |
|--------------|
| 1264761.96   |

Total revenue from all 1,200 orders was £1,264,761.96

#### Query 5 — Average Order Value

```
25  --- Query 5: What is the average amount spent per order ?
26  SELECT ROUND(AVG(TotalPrice), 2) AS AverageOrderValue
27  FROM EcommerceData
28
29  --- Query 6: Which product generates the most revenue?
30  SELECT Product,
31  ROUND(SUM(TotalPrice), 2) AS TotalRevenue
32  FROM EcommerceData
33  GROUP BY Product
```

0 % ▼

✓ No issues found

Ln: 29, Ch: 1 (13)

Results Messages

| AverageOrderValue |
|-------------------|
| 1053.97           |

Average order value was £1,053.97

## Query 6 — Revenue by Product

```
28
29 --- Query 6: Which product generates the most revenue?
30 SELECT Product,
31 ROUND(SUM(TotalPrice), 2) AS TotalRevenue
32 FROM EcommerceData
33 GROUP BY Product
34 ORDER BY TotalRevenue DESC
35
36 --- Query 7: Which payment method do customer use the most?
37 SELECT PaymentMethod, COUNT(*) AS TotalOrders
38 FROM EcommerceData
39 GROUP BY PaymentMethod
```

0 % No issues found Ln: 34, Ch:

Results Messages

| Product | TotalRevenue |
|---------|--------------|
| Chair   | 195620.11    |
| Printer | 195612.61    |
| Laptop  | 192126.56    |
| Tablet  | 186568.95    |
| Monitor | 175651.41    |
| Desk    | 167459.93    |
| Phone   | 151722.39    |

Chair generated the highest revenue at £195,620.11 despite Printer having the most orders.

## Query 7 — Payment Method

```
35
36 --- Query 7: Which payment method do customer use the most?
37 SELECT PaymentMethod, COUNT(*) AS TotalOrders
38 FROM EcommerceData
39 GROUP BY PaymentMethod
40 ORDER BY TotalOrders
41
42 --- Query 8: What is the distribution of order statuses?
43 SELECT OrderStatus, COUNT(*) AS TotalOrders
44 FROM EcommerceData
45 GROUP BY OrderStatus
```

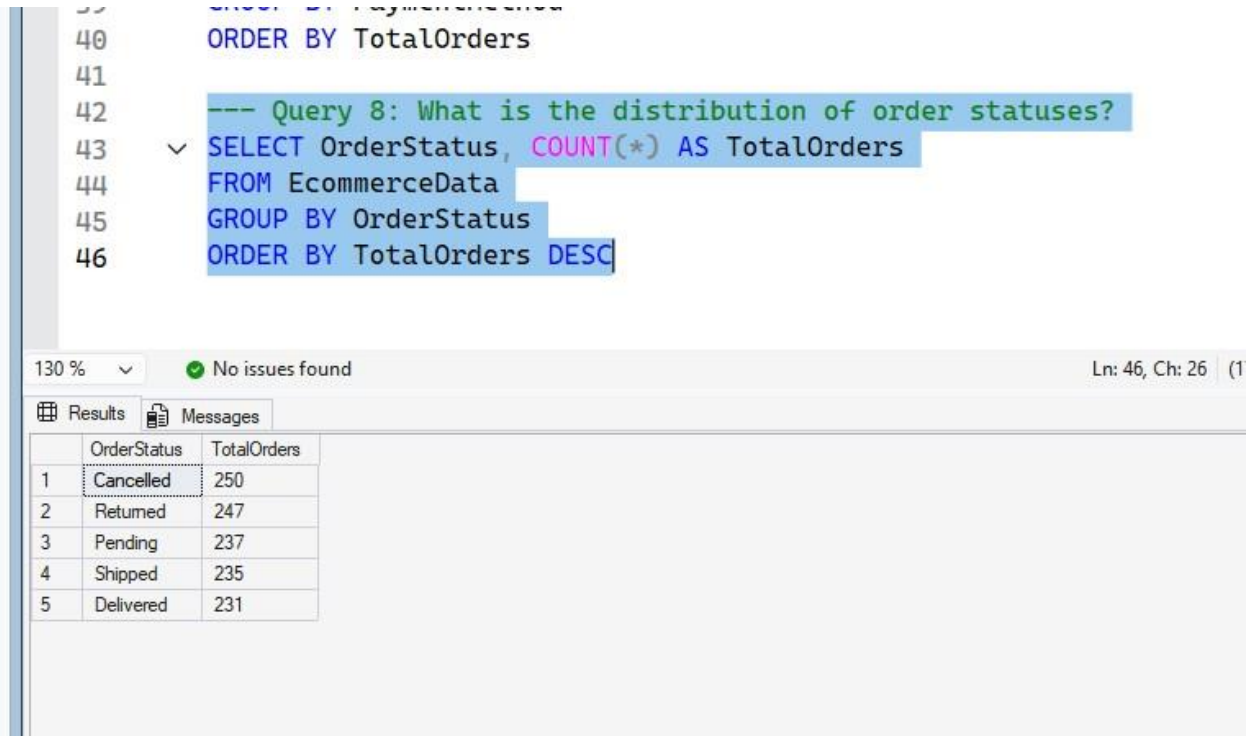
130 % No issues found Ln: 42, Ch: 1 (177 cha

Results Messages

|   | PaymentMethod | TotalOrders |
|---|---------------|-------------|
| 1 | Gift Card     | 230         |
| 2 | Debit Card    | 232         |
| 3 | Credit Card   | 234         |
| 4 | Cash          | 246         |
| 5 | Online        | 258         |

Online payment was most preferred with 258 orders.

#### Query 8 — Order Status



```
40 ORDER BY TotalOrders
41
42 --- Query 8: What is the distribution of order statuses?
43 SELECT OrderStatus, COUNT(*) AS TotalOrders
44 FROM EcommerceData
45 GROUP BY OrderStatus
46 ORDER BY TotalOrders DESC
```

130 % No issues found Ln: 46, Ch: 26 (1)

Results Messages

|   | OrderStatus | TotalOrders |
|---|-------------|-------------|
| 1 | Cancelled   | 250         |
| 2 | Returned    | 247         |
| 3 | Pending     | 237         |
| 4 | Shipped     | 235         |
| 5 | Delivered   | 231         |

Cancelled orders were highest at 250 — representing a major business concern.

### 3. Key Findings

41% of orders were either cancelled or returned

Chair generates most revenue despite not being most ordered

Online payment is most preferred by customers

Only 19% of orders were successfully delivered

### 4. Recommendations

Investigate high cancellation and return rates urgently

Promote Phone products to boost sales

Improve delivery process to increase successful deliveries.

## **5. Conclusion**

The SQL analysis revealed critical business insights particularly around order cancellations that require immediate attention.